Lesson 9.5 - How to Calculate the Sample Size

> **Video:** How To Calculate The Sample Size Given The Confidence Level & Margin of Error by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Formulas

For a mean. Start from the margin of error:

$$E = z\,\frac{\sigma}{\sqrt{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\sigma^2}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\sigma^2}{E^2}}$$

For a proportion. Same algebra, starting from the proportion's margin of error (Lesson 9.4):

$$E = z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\,\hat{p}(1-\hat{p})}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\,\hat{p}\,(1-\hat{p})}{E^2}}$$

It is the mean formula with σ^2 replaced by $\hat{p}(1-\hat{p})$, as explained below.

If $\hat{p}$ is unknown, use $\hat{p} = 0.5$. That makes $\hat{p}(1-\hat{p})$ as large as possible, so the resulting n is big enough whatever the true proportion is.

Always **round** n **up** to the next whole number.

The logic behind the formulas

1. Every sample gives a different answer. Survey n people and you get a sample proportion $\hat{p}$. A different random group would give a slightly different $\hat{p}$. Across many samples, the $\hat{p}$ values form a bell curve centred on the true p. The spread of that curve is the **standard error**:

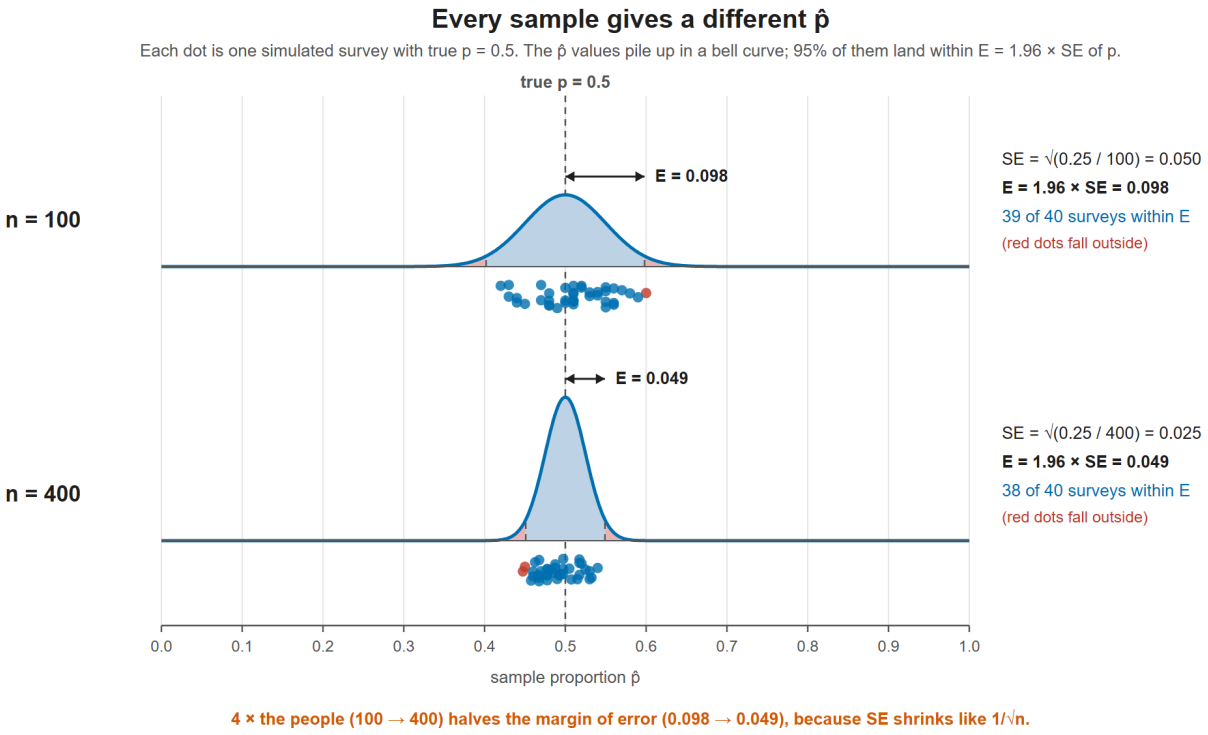
$$SE = \sqrt{\frac{p(1-p)}{n}}$$

Why p here and not $\hat{p}$? The SE describes how much $\hat{p}$ wobbles from sample to sample. That wobble comes from the population the samples are drawn from, so the exact formula uses the true p. But p is exactly what we don't know. In practice we plug in the best stand-in available: the sample's $\hat{p}$ when building an interval from data (Lesson 9.4), or a guess, or 0.5, when planning a survey before any data exists (point 6). That swap is why the formulas at the top are written with $\hat{p}$. With a decent-sized sample $\hat{p}$ is close to p, so the estimated SE is close to the true one.

2. The margin of error is z standard errors. On a bell curve, 95% of values lie within 1.96 standard deviations of the centre. So 95% of the time, $\hat{p}$ lands within $1.96 \times SE$ of the truth:

$$E = z \times SE$$

E is how far off you are willing to be, and z is how sure you want to be of staying within that distance.



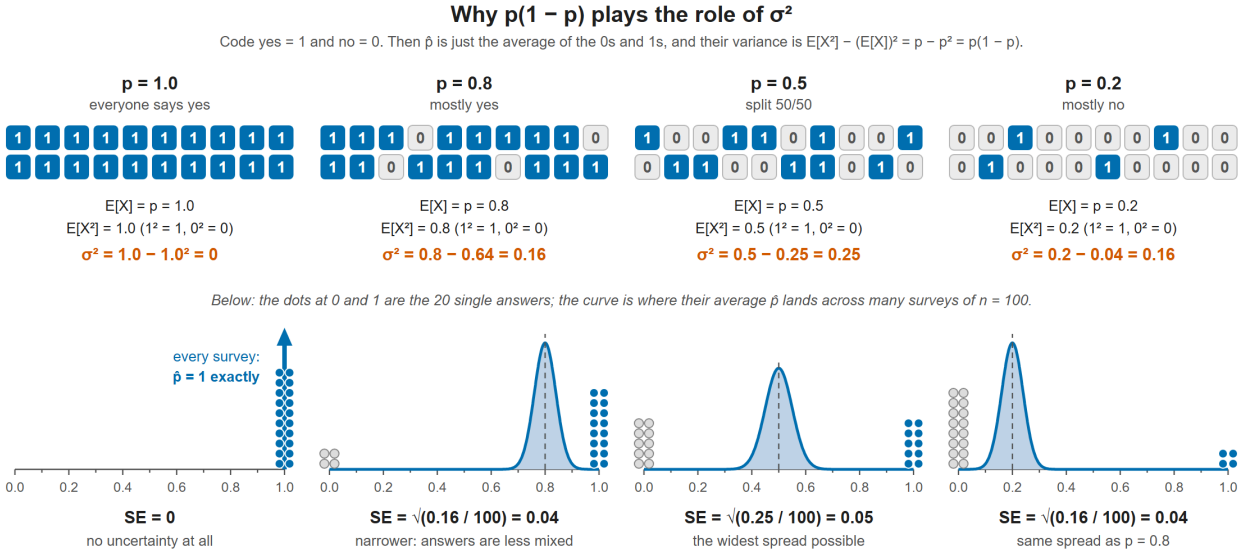
Each dot is one simulated survey. Its $\hat{p}$ misses the true $p = 0.5$ by a different amount each time, but about 95% of the dots land inside $\pm E$. The bottom row previews point 4: four times as many people makes the curve half as wide.

3. Why $p(1-p)$ plays the role of σ^2. Code each answer as 1 (yes) or 0 (no). Then $\hat{p}$ is just the average of those 0s and 1s, and their variance is

$$\sigma^2 = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

since $1^2 = 1$ and $0^2 = 0$. Intuitively:

- If everyone says yes ($p = 1$), every sample looks the same: $p(1-p) = 0$, so there is no uncertainty.
- If the population is split 50/50, samples can easily tilt either way, so the variation is largest: $p(1-p) = 0.25$.

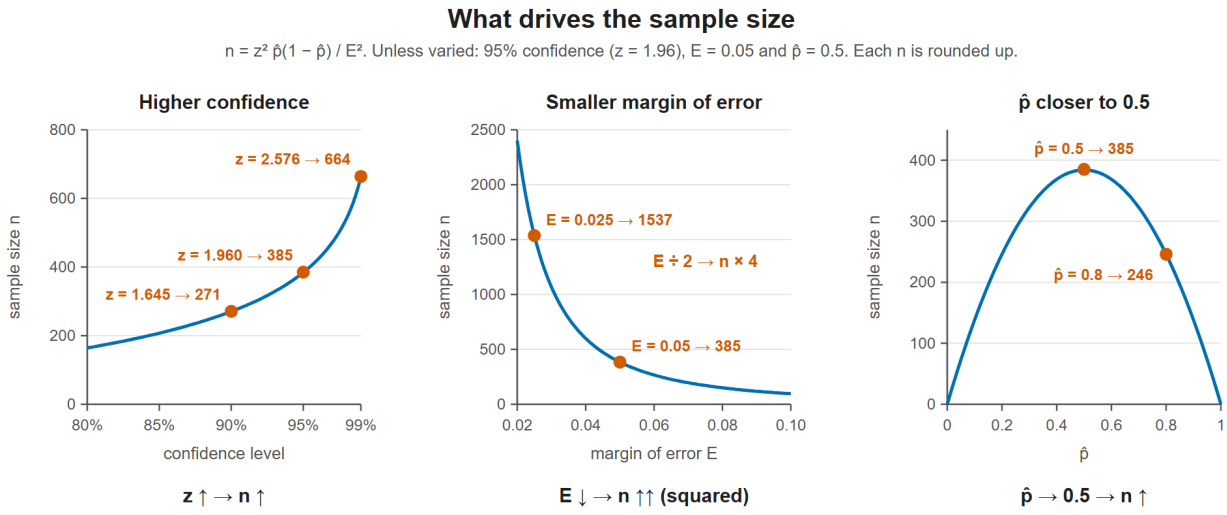


A single answer can only be 0 or 1 (the dots), but the average of 100 answers clusters tightly around p (the curve). The more mixed the 0s and 1s, the bigger $p(1-p)$, and the wider $\hat{p}$ spreads from survey to survey. $p = 0.2$ and $p = 0.8$ give the same spread, since $0.2 \times 0.8 = 0.8 \times 0.2$: only how mixed the answers are matters, not which side is the majority.

4. Why divide by n. In a small sample, a few unusual answers move $\hat{p}$ a lot. In a large one they get diluted. The variance shrinks like $1/n$, so SE shrinks like $1/\sqrt{n}$. Because of the square root, **halving** the margin of error takes **4 times** as many people.

5. Turn it around. Normally you collect data and then compute E. Sample-size planning runs the other way: you choose the E you want and solve for the n that achieves it.

Change	Effect on n	Why
Higher confidence (z ↑)	n ↑	Being more sure needs more evidence
Smaller margin of error (E ↓)	n ↑↑ (squared)	More precision needs more data
p closer to 0.5	n ↑	A more mixed population is less predictable



Each panel changes one ingredient and holds the others at 95%, $E = 0.05$, $\hat{p} = 0.5$. The middle panel is the steepest: halving E multiplies n by 4.

6. Why $\hat{p} = 0.5$ is the safe choice. You plan the survey before you have data, so p is unknown. The safe move is to use the $\hat{p}$ that makes n as large as possible. Since z and E are fixed, that means finding the maximum of

$$f(\hat{p}) = \hat{p}(1-\hat{p}) = \hat{p} - \hat{p}^2, \qquad 0 \le \hat{p} \le 1$$

With calculus. Set the first derivative to zero:

$$f'(\hat{p}) = 1 - 2\hat{p} = 0 \;\Rightarrow\; \hat{p} = 0.5$$

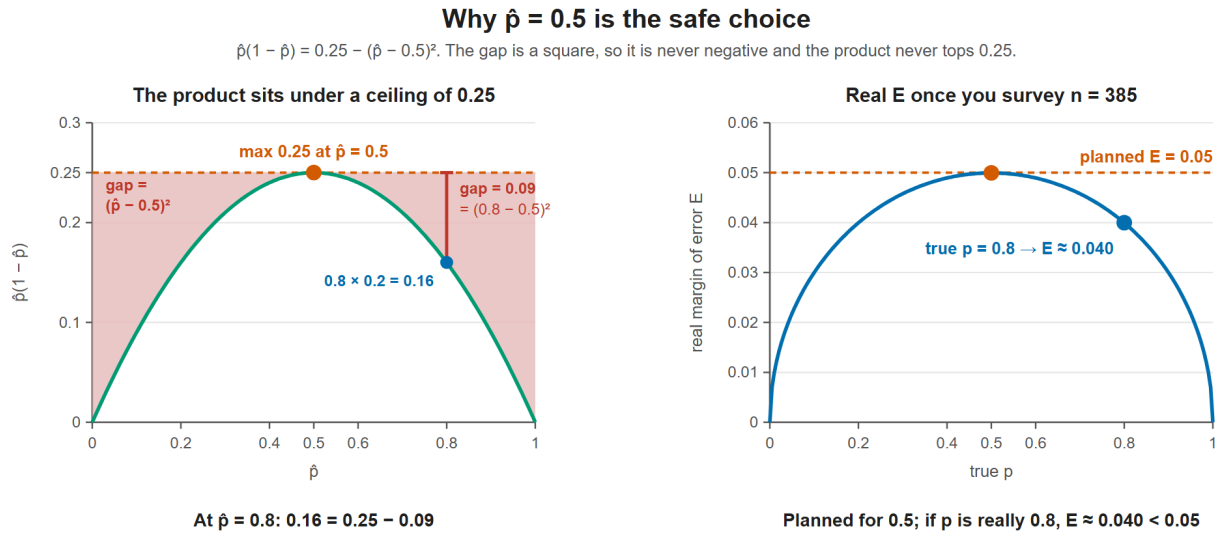
The second derivative is $f''(\hat{p}) = -2 < 0$, so the curve bends downward everywhere and $\hat{p} = 0.5$ is a maximum, not a minimum. Its value is $f(0.5) = 0.5 \times 0.5 = 0.25$. At the ends, $f(0) = f(1) = 0$, so 0.25 is the largest value on the whole range $0 \le \hat{p} \le 1$.

Without calculus. Completing the square gives the same answer:

$$\hat{p}(1-\hat{p}) = 0.25 - (\hat{p} - 0.5)^2$$

The squared term is never negative, so the product can never exceed 0.25, which happens only at $\hat{p} = 0.5$.

Planning for this worst case guarantees n is large enough. If the true p turns out to be, say, 0.8, the real margin of error simply comes out a bit smaller than planned.



Left: the shaded gap between the curve and 0.25 is $(\hat{p} - 0.5)^2$, which is zero only at $\hat{p} = 0.5$. Right: survey the planned $n = 385$, and if p turns out to be 0.8 the margin of error is $1.96\sqrt{0.16/385} \approx 0.040$, comfortably inside the 0.05 you asked for.

> **In one sentence:** sample answers wobble around the truth. The wobble grows with how mixed the population is and shrinks with $\sqrt{n}$, so you pick n large enough that z times the wobble fits inside the error you'll accept.

Example 1 - working backwards to the class size

The mean test score was 84 with standard deviation $\sigma = 15$. The margin of error at 95% confidence was $E = 5.367$. How many students were sampled?

$$n = \frac{(1.96)^2(15)^2}{(5.367)^2} \approx 30.0 \;\Rightarrow\; n = 30 \text{ students}$$

(The mean, 84, isn't needed.)

Example 2 - planning a survey on a proportion

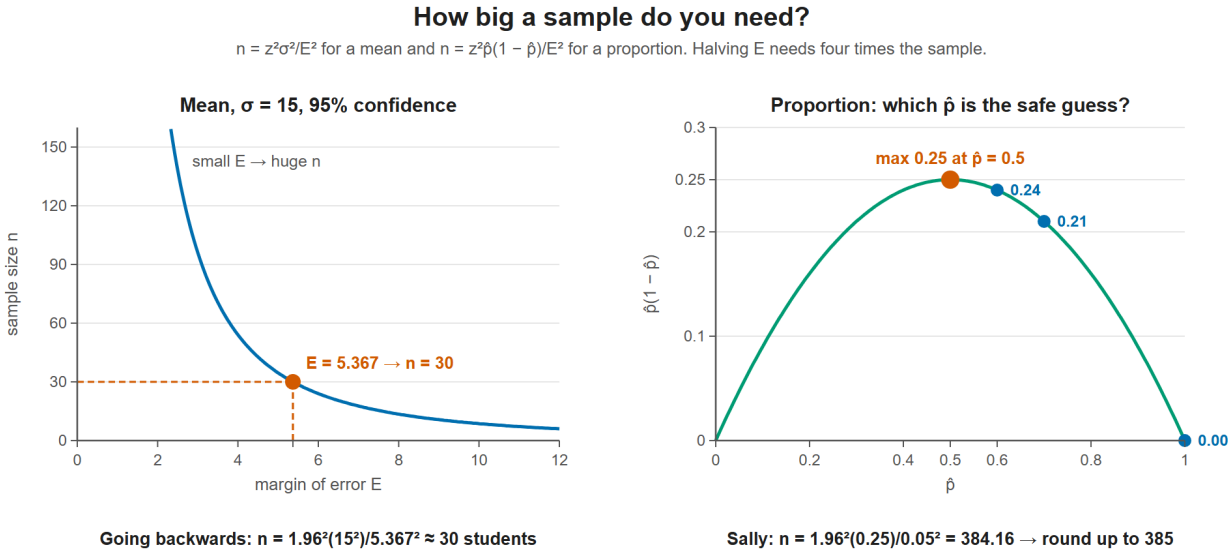
Sally wants to estimate the proportion of high-school students planning to go to college, with **95%** confidence and a **5%** margin of error ($E = 0.05$). She has no prior estimate of $\hat{p}$.

Which $\hat{p}$ makes $\hat{p}(1 - \hat{p})$ largest?

$\hat{p}$	$1-\hat{p}$	Product
1.0	0.0	0.00
0.7	0.3	0.21
0.6	0.4	0.24
0.5	**0.5**	**0.25** (largest)

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2} = \frac{0.9604}{0.0025} = 384.16 \;\Rightarrow\; n = 385$$

Round **up**: 384 students would not quite reach the 5% margin.



Left: the required n grows fast as the margin of error shrinks, and Example 1's E = 5.367 corresponds to n = 30. Right: $\hat{p}(1-\hat{p})$ is largest at $\hat{p} = 0.5$, so using 0.5 when nothing is known gives the safest (largest) sample size.

Key takeaways

- Mean: $n = z^2\sigma^2/E^2$. Proportion: $n = z^2\hat{p}(1-\hat{p})/E^2$.
- Both come from solving $E = z \times SE$ for n. For yes/no data the variance is $p(1-p)$, which takes the place of σ^2.
- With no estimate of $\hat{p}$, use 0.5. This gives the largest, safest n.
- Always round **up**.
- A smaller margin of error or a higher confidence level requires a larger sample.